

Welcome to Statistical Reasoning

STAT UN 1001 • Lecture 1

Tian Zheng

2026-09-08

Our first data collection

How easy or difficult was this classroom to find?

Show a number with your fingers:

Thumbs up	no movement	Thumbs down
Very easy	In between	Very difficult

Why collect these data?

The responses can help me:

- learn whether students had trouble finding this room
- identify what made the trip difficult
- decide whether future students need clearer directions

What else would I need to know before choosing a solution?

Individual values and a summary statistic

The full dataset keeps every student's response:

A summary statistic compresses those values into one number. For example:

- the median difficulty rating
- the proportion of students who answered 4 or 5

What does each summary preserve, and what does it hide?

How far can today's data travel?

These responses directly describe the students who answered today.

They may help me plan for another class in this room, but the next situation could differ:

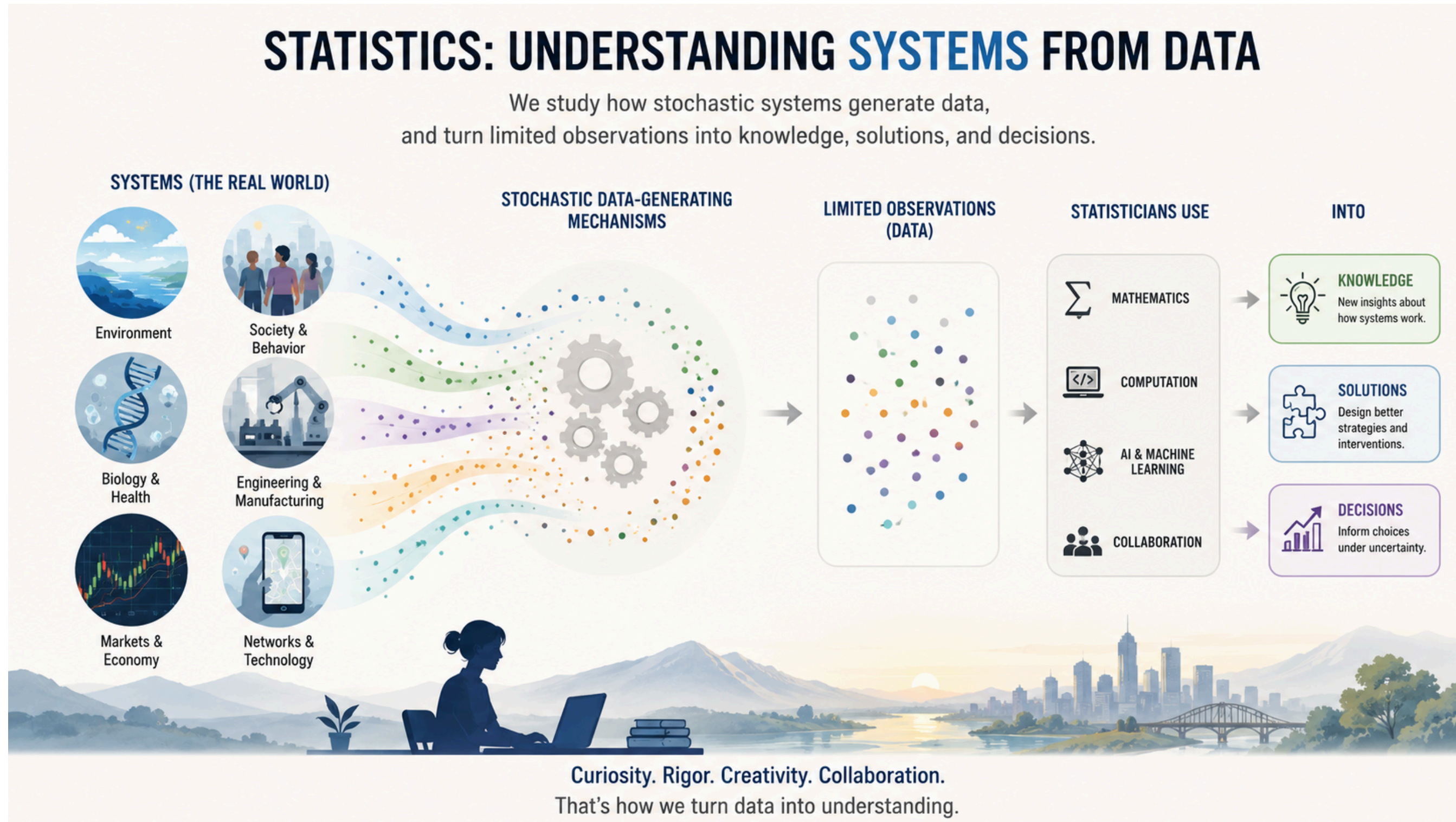
- different students or prior familiarity with the building
- different directions, entrances, or construction
- a different room, building, or time of day

What would make future data more informative?

Today

1. What statisticians do
2. How this course works
3. A study about ChatGPT and the brain
4. The basic language of data

Statistics: understanding systems from data



What does a statistician do?

Statisticians learn about a larger system from limited observations.

They ask:

- What exactly do the data represent?
- How were the observations produced?
- What patterns could arise by chance?
- Which conclusions does the evidence support?

My current research

Research question: Can we combine climate observations, physical models, and machine learning to improve near-term predictions of extreme rainfall?

The system: Earth's climate and the processes that produce heavy rain.

The observations: Weather stations, satellites, and climate-model simulations.

The decision or impact: Better estimates could help communities prepare for flooding.

This course in one sentence

We will learn how to make careful claims from imperfect data.

By the end of the semester, you should be able to:

- describe what data show
- recognize weak or overclaimed evidence
- reason about variability, uncertainty, and chance
- carry out and communicate a small data investigation

CourseWorks

Use CourseWorks to find:

- the weekly module and preparation
- slides, activities, and datasets
- assignment instructions and due dates
- announcements and course updates

For this week: open the Week 01 module.

The weekly rhythm

Before class	Tuesday	Thursday
Short preparation, usually due Monday evening	Concepts, questions, and worked examples	Activities, applications, project work, or a quiz
Flipped activities receive completion credit. Most mini-project work happens in class.		

A few syllabus essentials

- Read announcements and check the weekly CourseWorks module.
- Quizzes and the final exam are paper based.
- You may use approved AI tools as assistants, but you remain responsible for your reasoning and final work.
- Never enter private or confidential information into an AI tool.

Questions about the syllabus?

Course entry survey

Please complete the course entry survey.

The survey helps us understand the experiences and needs in the room.

A striking claim about ChatGPT

| Does using ChatGPT to write an essay harm your brain?

Before reading further:

1. What would “harm” have to mean?
2. What evidence would convince you?

What the study compared

The study placed participants in one of three essay-writing conditions:

- ChatGPT
- search engine
- no digital tools

Researchers measured EEG connectivity, features of the essays, recall, and participants' reported sense of ownership.

Cases, variables, and values

Term	Meaning	In this study
Case	One entity described by the data	One participant
Variable	A characteristic recorded for each case	Writing condition, EEG measure, recall score
Value	A variable's result for one case	"ChatGPT condition" or a particular score

Explanatory and response variables

Explanatory variable

Writing condition: ChatGPT, search engine, or no digital tools

Response variables

Outcomes that might change across conditions, such as EEG connectivity, recall, and essay features

Question: Which response best represents learning?

Sample and population

Sample: the participants who took part in this study

Population: the broader group we hope to understand

The sample included 54 participants and one kind of task: essay writing.

How broadly should we generalize?

From data to a claim

1. Start with a question about a population or process.
2. Observe a sample and define the variables.
3. Look for patterns while accounting for variability and study design.
4. State a conclusion that matches the evidence.

Statistics helps us judge the distance between step 2 and step 4.

Headline versus evidence

The evidence came from:

- a small sample
- a narrow writing task
- several direct and indirect outcomes
- a preprint that had not yet completed peer review

Rewrite the headline so that it says only what this study directly supports.

Our first statistical habit

When you encounter a data-based claim, ask:

Who or what are the cases?

Which variables were measured, and how?

What population does the sample represent?

Does the claim match the evidence?

Before Thursday

Bring your laptop to class.

We will experiment with AI together and practice checking what it gives us.

Also:

- complete the course entry survey
- review the syllabus
- check the Week 01 CourseWorks module